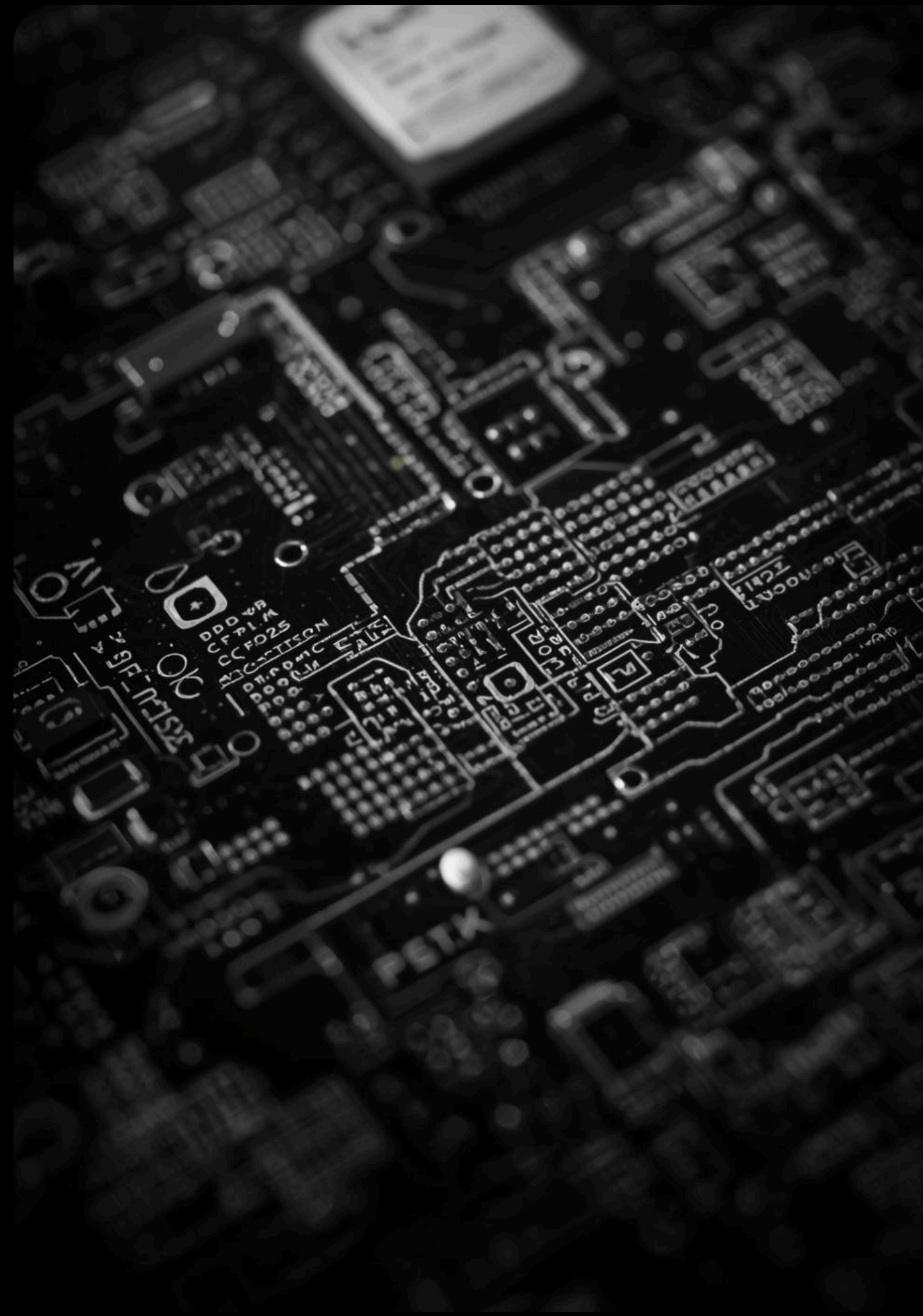


Day 10: Core AI Domains & Industry Use Cases



Why Day 10 Exists

So far, you have learned:

- ML fundamentals, EDA, modeling, evaluation
- Validation, overfitting, leakage
- Model selection and cost/architecture trade-offs

Now we zoom out and answer:

- Where does each AI technique actually fit in industry?

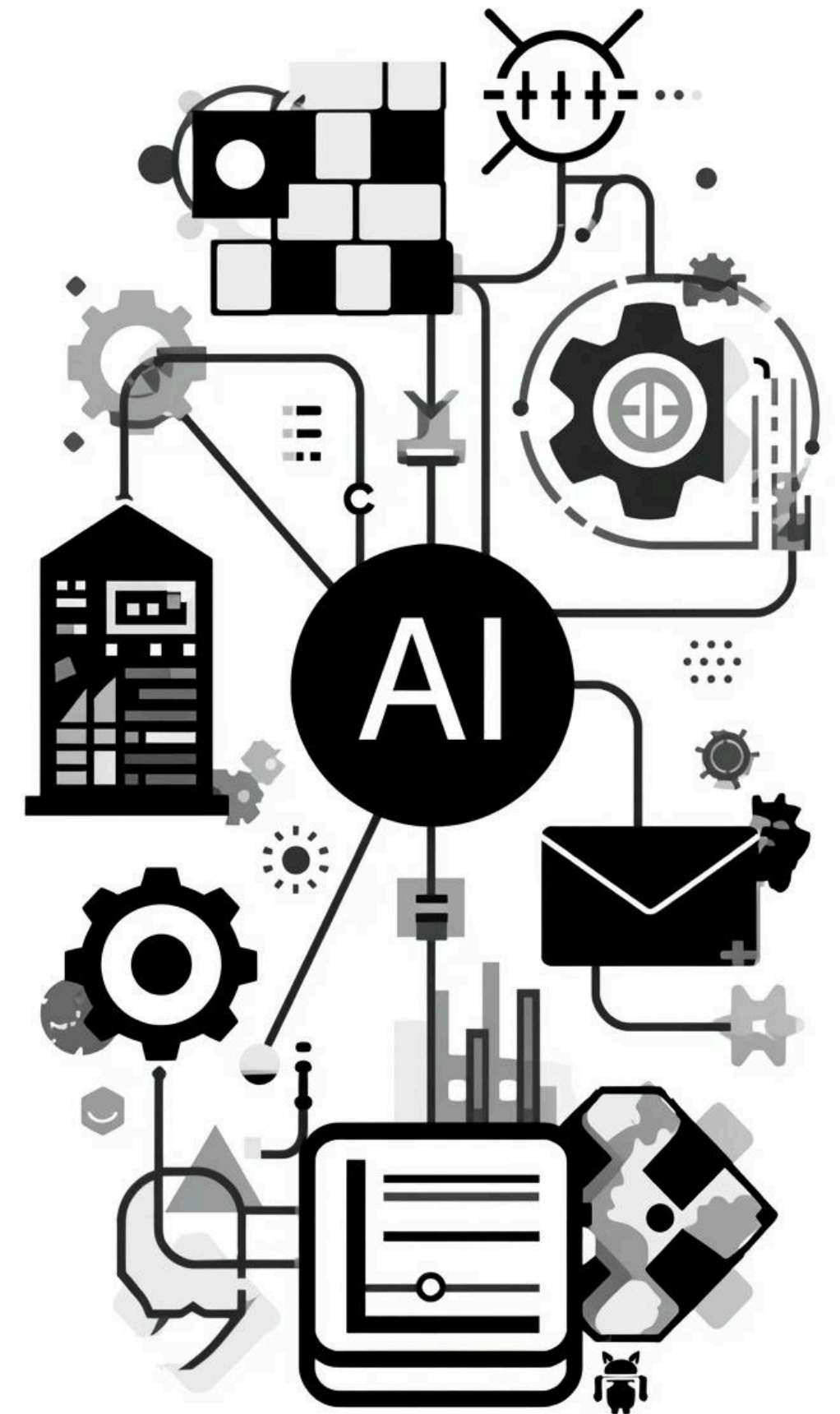


The Big Picture

Most real-world AI systems fall into one or more of these domains:

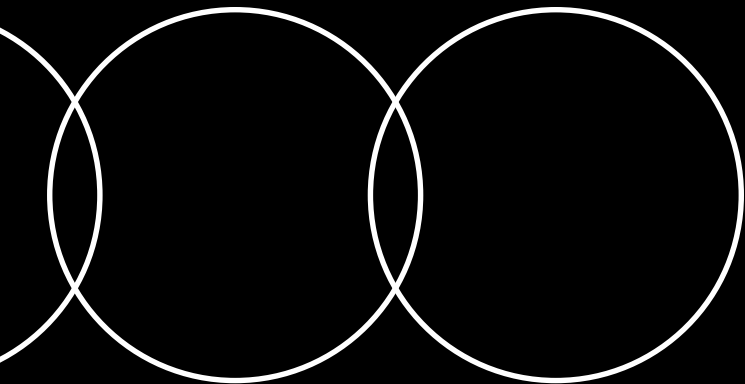
1. Computer Vision
2. Natural Language Processing (NLP)
3. Time Series Forecasting
4. Regression & Classification (Tabular ML)
5. Large Language Models (LLMs)

Understanding the domain determines the solution.



What Is Computer Vision?

Computer Vision enables machines to understand visual data



Understand Images

Computer Vision allows machines to interpret and analyze images, enabling them to recognize and understand visual content effectively.

Detect Objects

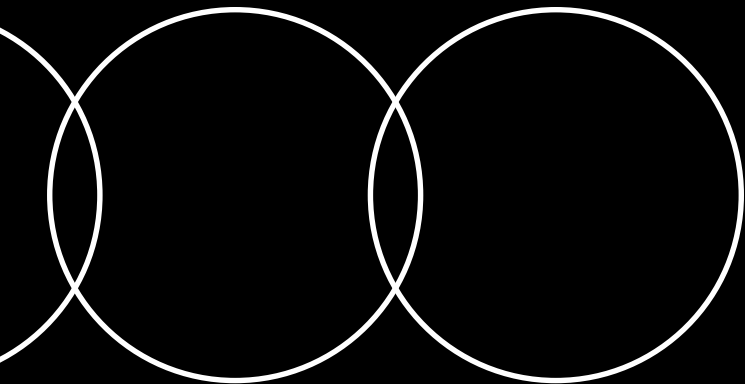
It empowers machines to detect various objects, patterns, and text within images or videos, enhancing their decision-making capabilities.

Make Decisions

By analyzing visual data, machines can make informed decisions based on the information derived from images and videos.

Image Classification

Understanding how to label images accurately



Task

Assign a label to an image, enabling machines to categorize and analyze visual data effectively in various applications.

Examples

Defective vs non-defective product and tumor vs non-tumor scan are common classifications performed in image analysis tasks.

Dataset example:

<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>

Common models

CNNs and transfer learning techniques like ResNet and EfficientNet are commonly used to enhance image classification performance.

Object Detection

- **Task:** Detect and localize objects in an image
- **Examples:** Detect vehicles in traffic footage, Count people in stores
- **Common models:** YOLO , Faster R-CNN
- **Dataset example:**
<https://www.kaggle.com/datasets/ultralytics/co128>



OCR & Document Processing

- **Task:** Extract text from images or scanned documents
- **Examples:** Invoice processing, ID card verification
- **Techniques:** OCR + NLP pipelines
- **Dataset / tools:** <https://github.com/tesseract-ocr/tesseract>
- **Industry use cases:** Finance, Government, Healthcare



CV Industry Use Cases

Industries using CV heavily:

- Manufacturing (quality inspection)
- Healthcare (medical imaging)
- Security & surveillance
- Retail (shelf monitoring)



What Is NLP?

NLP focuses on understanding text, extracting meaning, and structuring unstructured language. The primary data type is text, which forms the basis for various applications.

- Understanding text through context and semantics
- Extracting meaning from complex language structures
- Structuring unstructured language for better analysis

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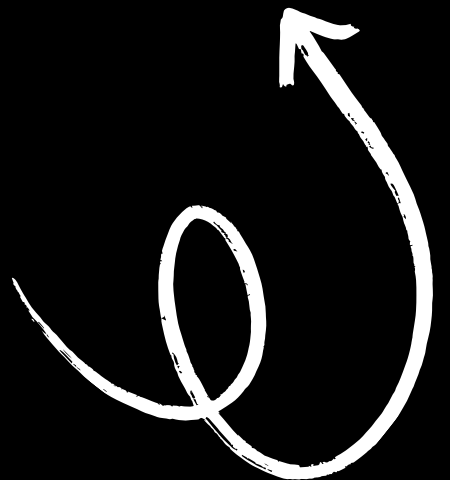
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Text Classification

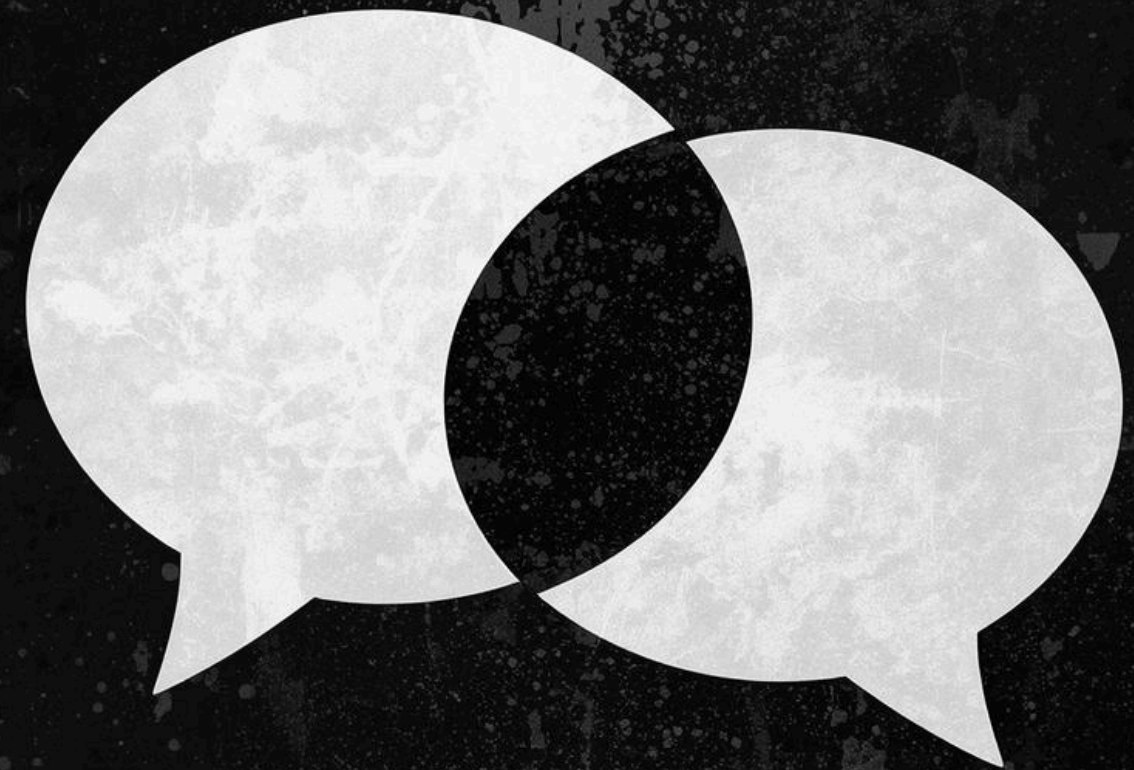
Assigning Categories to Text

The task of text classification involves **assigning categories** to textual data, enabling systems to automatically determine the meaning and context of the content provided.



Text Classification Examples

- **Sentiment analysis** identifies emotions behind the text provided
- **Spam detection** filters unwanted messages effectively in communications
- **Techniques:** TF-IDF + classical ML, Embeddings + classifiers
- **Dataset example:**
<https://www.kaggle.com/datasets/kazanova/sentiment140>

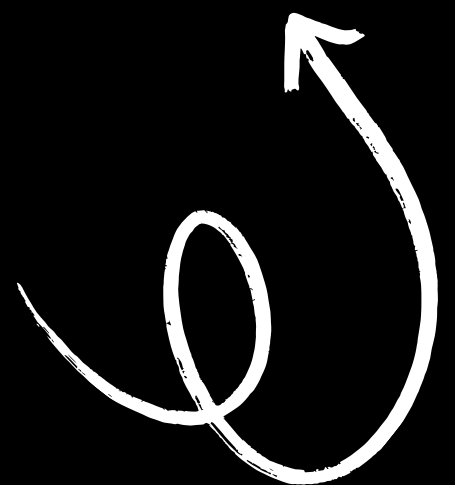


Information Extraction

Extracting structured data from text

The primary task of information extraction is to **extract structured data** from unstructured text. Examples include extracting fields from invoices and parsing resumes for relevant information.

- **Examples:** Invoice fields, Resume parsing
- **Techniques:** Named Entity Recognition (NER), Rule + ML pipelines
- **Dataset example:** <https://www.kaggle.com/datasets/dataturks/ner-dataset>



Search & Retrieval



Task

Find relevant documents effectively.



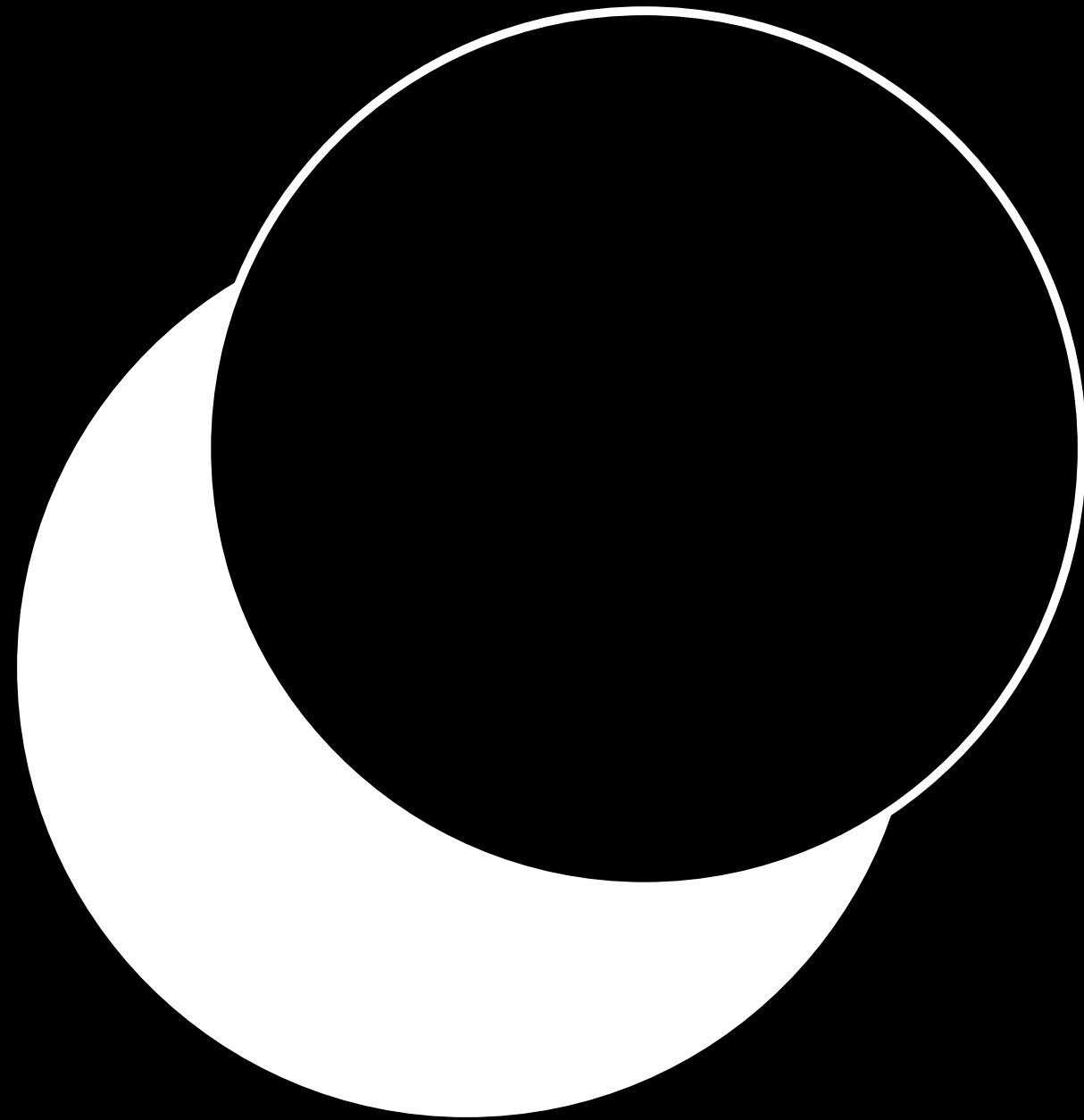
Examples

Enterprise search and knowledge base Q&A.



Techniques

Vector embeddings and similarity search methods.



Conversational AI and Chatbots Overview

Conversational AI enhances user experience through **customer support** and knowledge assistance using advanced techniques and systems like Intent classification (classical ML), LLM-based chatbots and RAG systems

What Is Time Series Data?



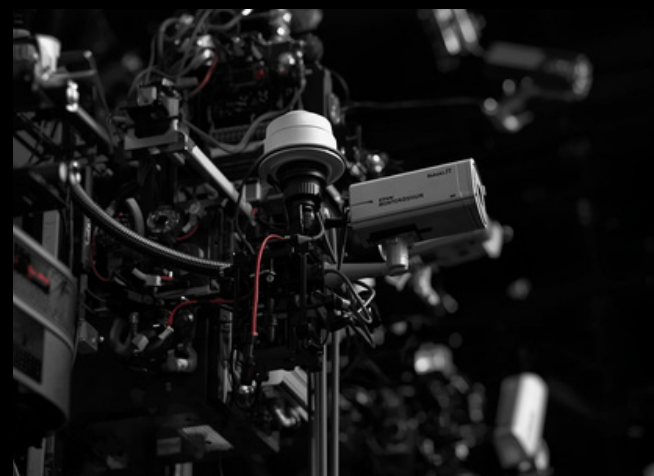
Indexed

Data is organized and classified by time.



Sales

Represents revenue changes tracked over specific periods.



Sensors

Collects data continuously for analysis and insights.



Examples

Illustrates various ways data can be represented.



Behavior

Indicates trends and patterns over time for analysis.

Forecasting Fundamentals: Key Concepts and Models



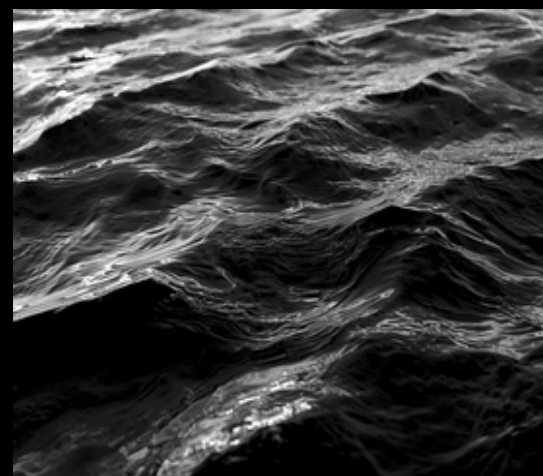
Trend

Identifies the **general direction** of data over time.



Seasonality

Captures **regular patterns** observed at specific intervals.



Noise

Refers to **random variability** affecting data clarity.



ARIMA

A **popular statistical method** for time series forecasting.



Prophet

An **open-source forecasting tool** developed by Facebook.

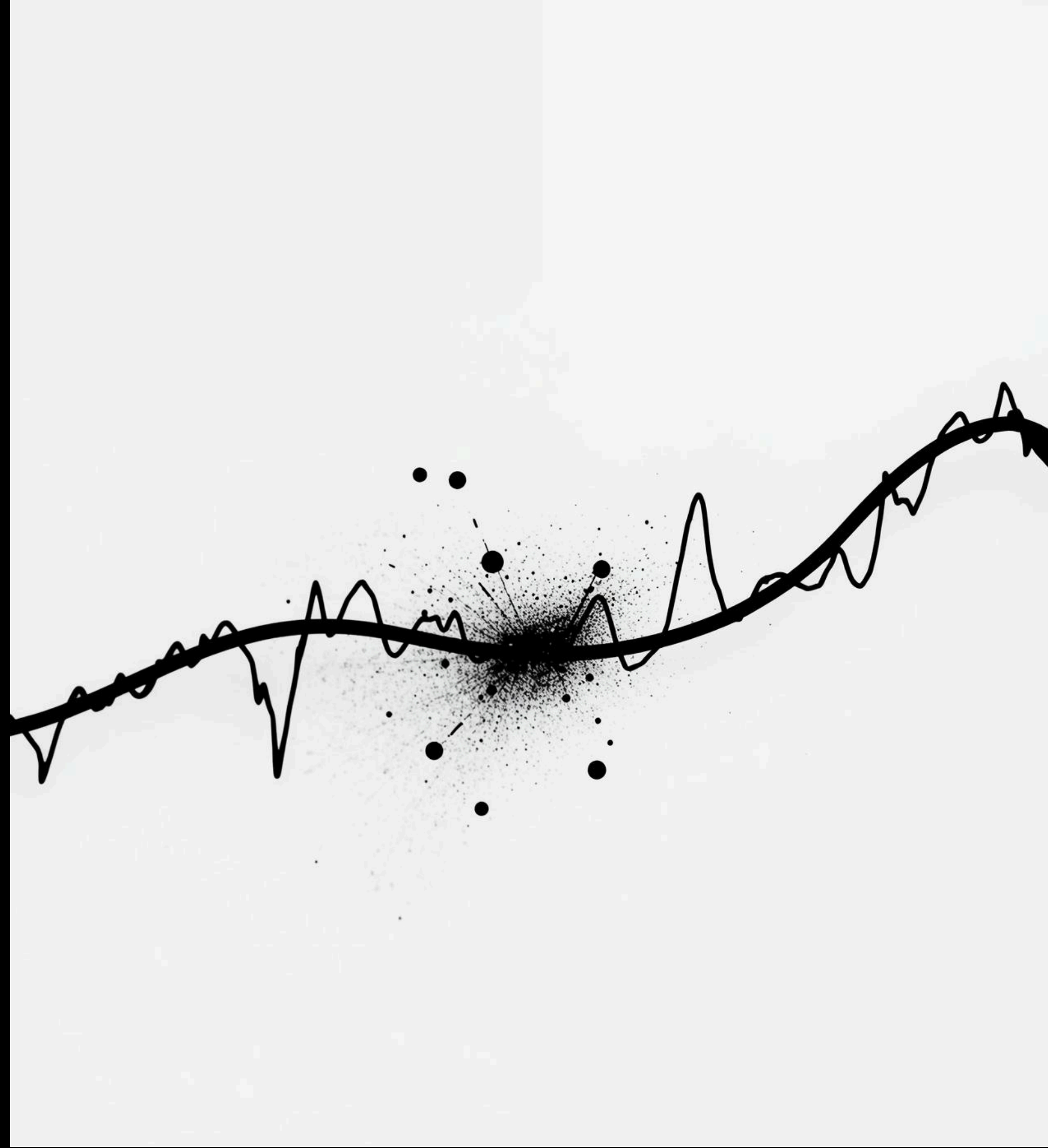


ML-based Regressors

Leveraging machine learning for advanced predictions.

Anomaly Detection Techniques

Anomaly detection focuses on identifying unusual behavior within data sets. This includes detecting instances of **fraud** and **system failures**, utilizing methods like statistical thresholds and machine learning-based detection techniques.



Use Cases

Finance

Risk and fraud detection

Retail

**Demand forecasting
applications**

Operations

**Predictive maintenance
strategies**

Healthcare

Patient monitoring systems

Regression in Industry Applications

Price Predictions

Regression models are essential for **predicting prices** in various industries, enabling businesses to set competitive pricing strategies based on market trends and consumer behavior.

Revenue Forecasting

By utilizing regression techniques, organizations can accurately forecast **revenue**, allowing for better financial planning and resource allocation, crucial for long-term success.

Demand Estimation

Regression analysis aids in estimating **demand** for products or services, helping companies optimize inventory levels and improve overall efficiency in supply chain management.

Classification in Industry Applications

Yes / No Outcomes

Classification is particularly useful for predicting **binary outcomes** such as determining whether a transaction is fraudulent or legitimate in various industries.

Categorization

This approach helps in segmenting data into distinct **categories**, enabling businesses to make informed decisions based on the classification of different customer behaviors.

Real-World Examples

Notable applications include **fraud detection** systems that minimize risk exposure and strategies addressing **customer churn** to improve retention rates and sales performance.

Feature Selection & Data Prep



Remove Leakage

Eliminate data that can skew results.



Handle Missing Values

Fill gaps to maintain dataset integrity.



Encode Categorical

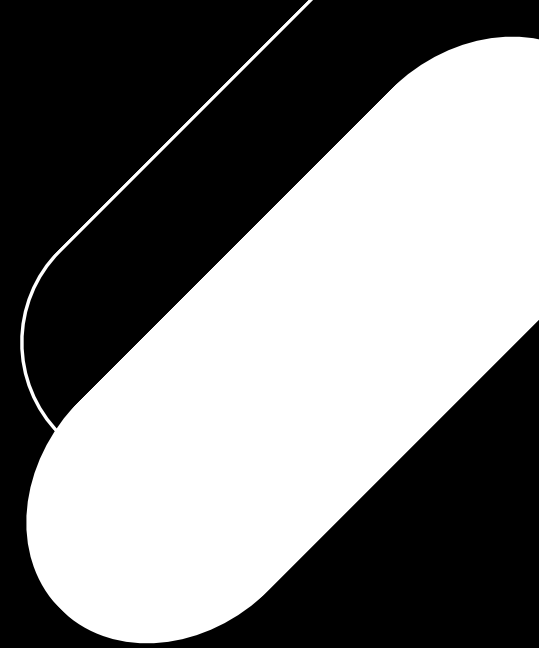
Transform non-numeric data for modeling.



Scale Numeric

Normalize features for effective learning.

Model Evaluation Metrics



Regression

Key metrics include **MAE**, **RMSE**, and **R²** to assess model performance in regression tasks.



Classification

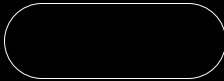
Important metrics such as **Precision**, **Recall**, and **F1** measure performance in classification problems effectively.



Business Alignment

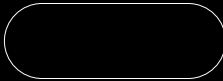
Metrics must align with **business impact** to ensure relevance and effectiveness in real-world applications.

Common Real-World Pitfalls



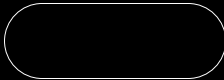
Wrong Domain

Using the wrong AI domain can lead to ineffective solutions and wasted resources in project development.



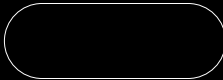
Overfitting

Overfitting occurs when a model learns noise instead of the underlying pattern, resulting in poor performance on new data.



Cost & Latency

Ignoring cost and latency considerations can significantly impact the feasibility and scalability of AI implementations.



Choosing LLMs

Choosing large language models (LLMs) unnecessarily can lead to inefficiencies and may not address specific business needs.

What Are Large Language Models?

Large Neural Networks

LLMs are large neural networks trained on extensive text corpora, enabling them to understand and generate human-like language effectively.

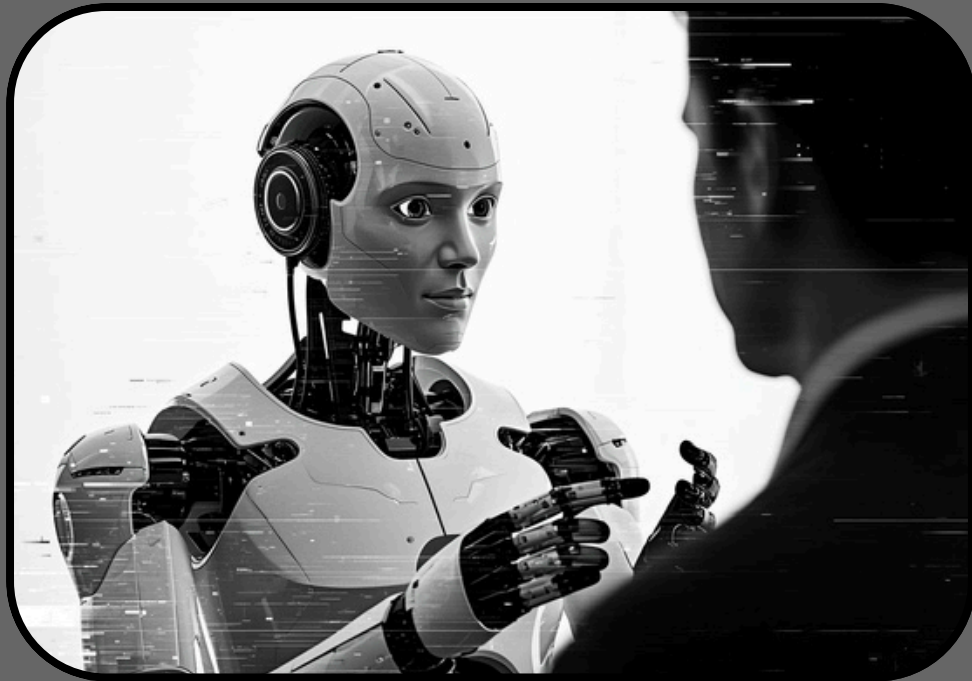
General-Purpose Language Models

These models serve as general-purpose tools for various applications, allowing flexibility in tackling a wide range of language-related tasks.

Notable Examples

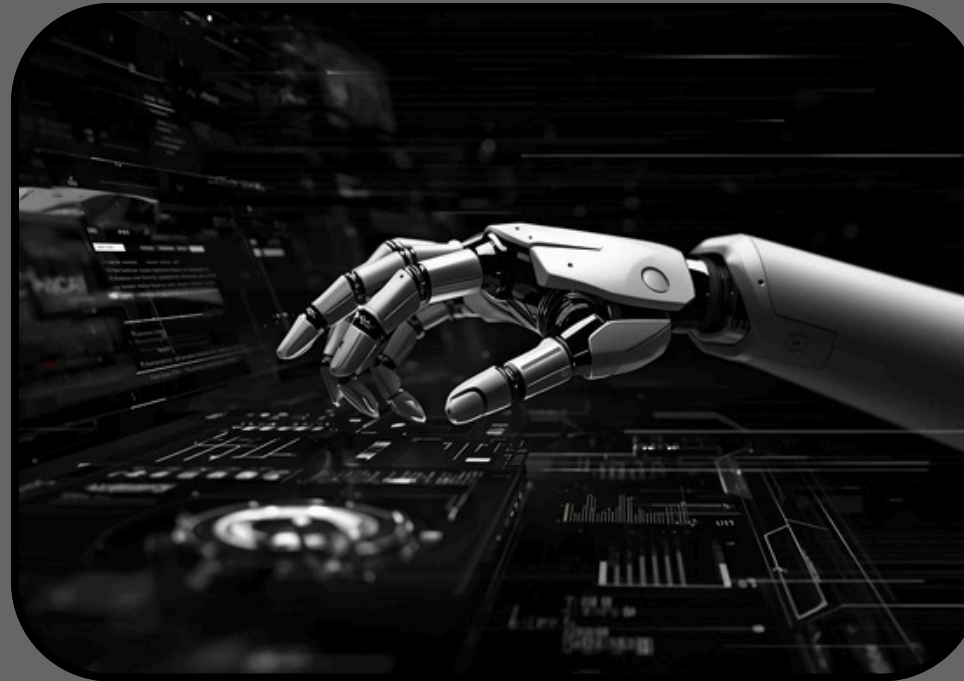
Examples of LLMs include GPT and LLaMA, both of which have demonstrated impressive capabilities in natural language processing and generation.

LLM Use Cases in Industry



Chatbots

Utilized for customer support and interaction.



Document summarization

Assists in condensing lengthy texts efficiently.



Code assistance

Aids developers in writing and debugging code.

When to Train or Fine-Tune a Model

Specific Domain Language

Train or fine-tune a model when the domain language is specific, ensuring the model understands unique terminology and nuances relevant to the particular field.

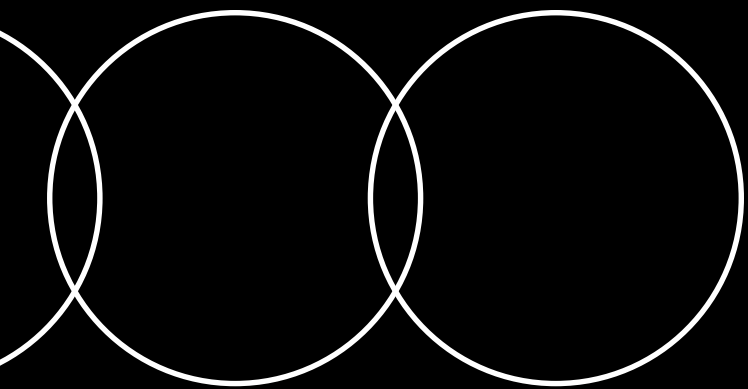
Off-the-Shelf Underperformance

If off-the-shelf models underperform for your needs, training or fine-tuning allows for better alignment with the specific requirements and complexities of your tasks.

Sufficient Labeled Data

Fine-tuning is beneficial when there is enough labeled data available, enabling the model to learn from real examples and improve its performance effectively.

End-to-End Case Study



Case Study Exercise

Choose a system to explore AI applications and techniques fully.

System Options



Invoice Processing

Automating invoicing can save **significant time** and effort.



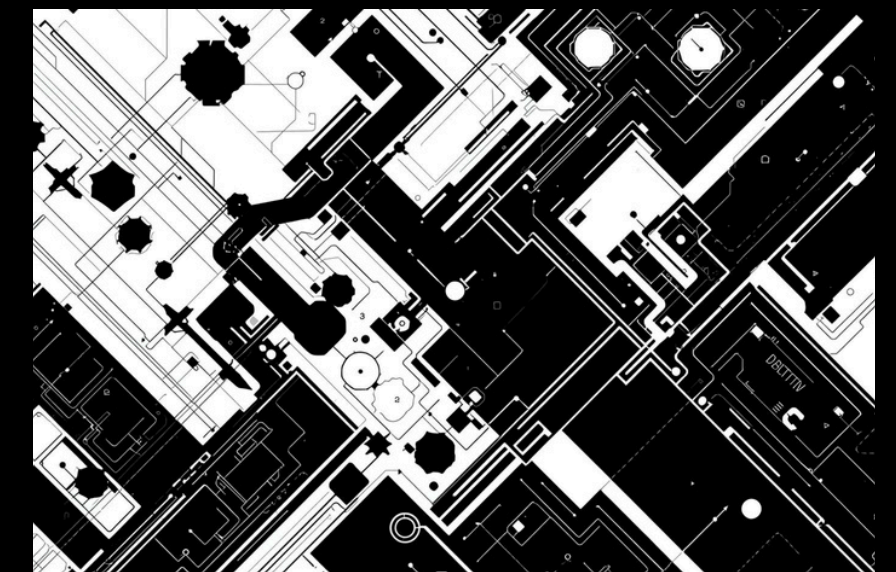
Customer Support

Chatbots enhance customer service by providing **instant responses**.



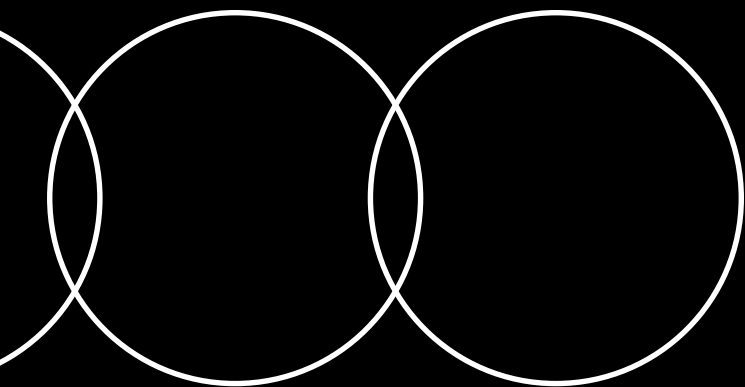
Sales Forecasting

Predicting trends helps organizations prepare for the **future demands**.



System Architecture

Understanding architecture is crucial for **effective implementation**.



For Your Case



Identify Domain

Focus on the specific area of application for AI.



Choose Technique

Select the appropriate method to address the identified problem.



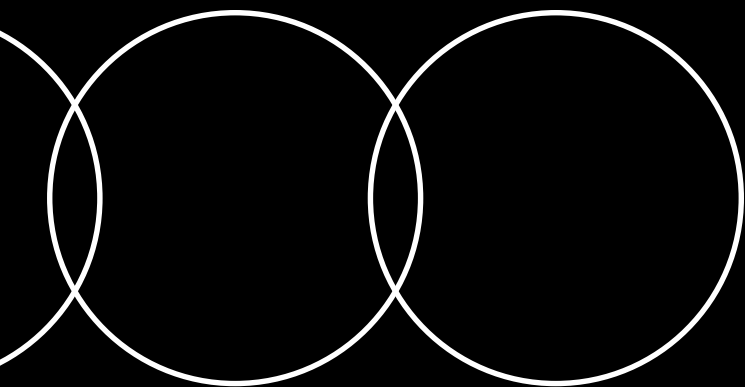
Define Metrics

Determine key performance indicators for evaluating success.

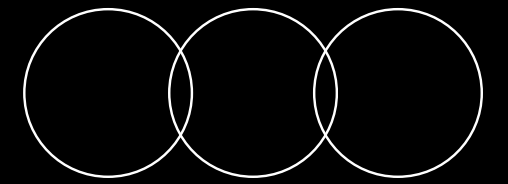


Sketch Architecture

Visualize the system architecture based on chosen techniques.



Hands-On Assessment



Industry Problem

Identify a significant challenge within your chosen industry to address.



Proposed Solution

Develop a clear and actionable plan to solve the identified industry problem.

Participants Must



Industry Problem

Choose a relevant issue to address in AI solutions.



AI Domain

Identify the AI domain that best fits your problem.



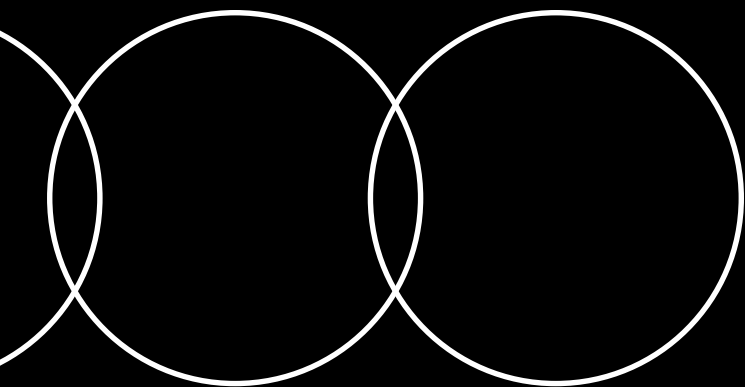
Proposed Solution

Formulate a solution tailored to the identified problem.



Justify Decisions

Provide rationale for your chosen approach and techniques.



Key Outcomes

By the end of Day 10, you should understand

AI Domain Fit

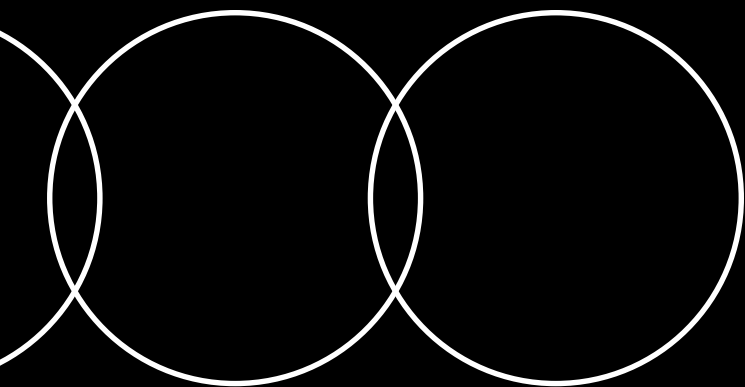
You will know where each AI domain fits within the context of real-world applications and industry challenges.

Mapping Techniques

Learn how to effectively map problems to the correct techniques that can provide solutions for those challenges.

Confidence in Use Cases

Speak confidently about various use cases, demonstrating your understanding and ability to apply AI concepts to real projects.



What Comes Next

Real Project Builds

Participants will engage in hands-on experience, applying their knowledge to develop real-world AI projects that address industry needs.

Interview Preparation

Participants will receive guidance on crafting effective resumes and preparing for interviews in the AI industry, enhancing their job readiness.

Case Study Development

Teams will create detailed case studies, documenting their AI solutions and methodologies to illustrate best practices and lessons learned.

Advanced LLM Workflows

The focus will shift towards more complex workflows involving large language models, emphasizing their application in various sectors.



Thank You